

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CE-702(D)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Structural Design and Drawing (RCC-II)

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. A six storey three bay building is to be located at Delhi. Determine lateral forces on an inner frame due to wind load and earthquake loads. Using the following data: 14
Bay Width = 6 m c/c
Frame Spacing = 4.5 m c/c
Height of ground floor = 4.2 m
Height of other floor = 3.7 m
Live Load = 5 kN/m²
Floor Thickness = 15 cm
Floor Finish = 4 cm
Columns = 30 cm × 60 cm
Beam = 30 cm × 40 cm
2. Design the stem of a cantilever retaining wall to retain an earth embankment with a horizontal top 3.75 m above ground level. Density of earth = 19 kN/m³, Angle of internal friction $\phi = 30^\circ$, SBC of soil = 180 kN/m². Coefficient of internal friction between soil and concrete = 0.5. Adopt M20 grade concrete and Fe 415 grade steel. 14

CE-702(D)-CBGS

PTO

[2]

3. Design a circular tank with flexible base for capacity of 4,00,000 liters. The depth of water is to be 4m, including a free board of 200 mm. Use M20 concrete and Fe 415 steel. Give details of reinforcement. 14
4. Explain the difference between Silo and Bunker. Also discuss the design of hopper bottom of a bunker. 14
5. A pretensioned prestress concrete beam of 7m span has cross section of 300 mm × 500 mm, is prestressed with 155 kN force at transfer. The cable has cross sectional area of 1500 mm² of steel and has a parabolic profile with a maximum eccentricity of 1500 mm at the mid span. Determine the loss of prestress, given that: $E_s = 2.1 \times 10^5 \text{ N/mm}^2$ and $E_c = 3 \times 10^4 \text{ N/mm}^2$. 14
6. Design a slab bridge with carriage way of 12 m with Kerbs only for an effective span of 6.0 m for IRC 70 loading. Concrete used is M25 and HYSD bar of grade Fe415 are used. 14
7.
 - a) Discuss various factors to be considered for determination of the general dimension of the rectangular shear walls. 7
 - b) What are the functions of retaining wall? Discuss the component of counterfort type retaining wall. 7
8. Write short notes on the following: 14
 - i) Types of prestressing
 - ii) IRC loads
 - iii) Post Tensioning System
